

```
1. #include<bits/stdc++.h>
2. using namespace std;
3.
4. int a[] = {1,0,-1,0,1};
5. int m,n;
6.
7. void markLand(vector<vector<int>> &A, int x,int y, int k){
8.     if(k == 0)
9.         return;
10.    for(int p=0; p<5; p++){
11.        int i = x+a[p];
12.        int j = y+a[p+1];
13.        if(i>=0 && j>=0 && i<m && j<n && A[i][j] != 1){
14.            A[i][j] = 2;
15.            markLand(A,i,j,k-1);
16.        }
17.    }
18. }
19.
20. int solution(vector<vector<int>> &A, int k){
21.    m = A.size();
22.    n = A[0].size();
23.
24.    for(int i=0; i<m; i++){
25.        for(int j=0; j<n; j++){
26.            if(A[i][j] == 1){
27.                markLand(A,i,j,k);
28.            }
29.        }
30.    }
31.    int count = 0;
32.    for(int i=0; i<m; i++){
33.        for(int j=0; j<n; j++){
34.            if(A[i][j] == 0)
35.                count++;
36.        }
37.    }
38.    return count;
39. }
40.
41. int main(){
42.    vector<vector<int>> A ={{0,0,0,0,0,0,0,0},{0,0,0,0,0,0,0,0},{0,0,1,0,0,0,0,0},{0,0,0,0,0,0,0,0},
43.    {0,0,0,0,0,0,0,0},{0,0,0,0,0,0,0,0},{0,0,0,0,0,0,1,0},{0,0,0,0,0,0,0,0}};
44.    int k = 2;
45.    cout<<solution(A,k);
46.
47.    return 0;
48. }
```